HILB^D OF AN AFFINE SCHEME IS REPRESENTABLE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 410-414 of sources/stacks-pic/pinned-original-source.tex; proof span is lines 416-513, the complete author proof (96 body lines).

Editorial note. Navigation stubs: 4 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 1 of queue_pic.json (pb=4020, stmt 410-414, proof 416-513).

Editorial note. stacks-pic wave-1 (pi01): fresh adjudication accept (2026-10-14); pool per standing queue hypercovering -> pic (SMC reassigned to Zhen per Chief 08:5x); pic.tex 27 labels/23 with proof judged 21 accept / 2 reject / 4 pb=0 env-dup rows excluded; spans cite-free and reference/slogan/history-free so no patchenv needed.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 00O0: sources/stacks-pic/pinned-original-source.tex, source lines 513–513. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0B97: sources/stacks-pic/pinned-original-source.tex, source lines 341–347. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0B95: sources/stacks-pic/pinned-original-source.tex, source lines 282–286. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01JJ: sources/stacks-pic/pinned-original-source.tex, source lines 513–513. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $X \to S$ be a morphism of affine schemes. Let $d \geq 0$. Then $\mathrm{Hilb}^d_{X/S}$ is representable.*

Date: stacks-project:pic.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. Say $S = \mathrm{Spec}(R)$. Then we can choose a closed immersion of X into the spectrum of $R[x_i; i \in I]$ for some set I (of sufficiently large cardinality. Hence by Lemma 0B97 we may assume that $X = \mathrm{Spec}(A)$ where $A = R[x_i; i \in I]$. We will use Schemes, Lemma 01JJ to prove the lemma in this case.

Condition (1) of the lemma follows from Lemma 0B95.

For every subset $W \subset A$ of cardinality d we will construct a subfunctor F_W of $\mathrm{Hilb}^d_{X/S}$. (It would be enough to consider the case where W consists of a collection of monomials in the x_i but we do not need this.) Namely, we will say that $Z \in \mathrm{Hilb}^d_{X/S}(T)$ is in $F_W(T)$ if and only if the $\mathcal{O}_T$-linear map

$$\bigoplus\nolimits_{f \in W} \mathcal{O}_T \longrightarrow (Z \to T)_*\mathcal{O}_Z, \quad (g_f) \longmapsto \sum g_f f|_Z$$

is surjective (equivalently an isomorphism). Here for $f \in A$ and $Z \in \mathrm{Hilb}^d_{X/S}(T)$ we denote $f|_Z$ the pullback of f by the morphism $Z \to X_T \to X$.

Openness, i.e., condition (2)(b) of the lemma. This follows from Algebra, Lemma 00O0.

Covering, i.e., condition (2)(c) of the lemma. Since

$$A \otimes_R \mathcal{O}_T = (X_T \to T)_*\mathcal{O}_{X_T} \to (Z \to T)_*\mathcal{O}_Z$$

is surjective and since $(Z \to T)_*\mathcal{O}_Z$ is finite locally free of rank d, for every point $t \in T$ we can find a finite subset $W \subset A$ of cardinality d whose images form a basis of the d-dimensional $\kappa(t)$-vector space $((Z \to T)_*\mathcal{O}_Z)_t \otimes_{\mathcal{O}_{T,t}} \kappa(t)$. By Nakayama's lemma there is an open neighbourhood $V \subset T$ of t such that $Z_V \in F_W(V)$.

Representable, i.e., condition (2)(a) of the lemma. Let $W \subset A$ have cardinality d. We claim that F_W is representable by an affine scheme over R. We will construct this affine scheme here, but we encourage the reader to think it through for themselves. Choose a numbering $f_1, \ldots, f_d$ of the elements of W. We will construct a universal element $Z_{univ} = \mathrm{Spec}(B_{univ})$ of F_W over $T_{univ} = \mathrm{Spec}(R_{univ})$ as follows. First, we set

$$R_{univ} = R[c^m_{kl}, b^l, b^l_i]/\mathfrak{a}_{univ}$$

for some ideal $\mathfrak{a}_{univ}$ which we describe below. Here the indices k, l, m, i are all ranging through $\{1, \ldots, d\}$. Then we set

$$B_{univ} = R_{univ}e_1 \oplus R_{univ}e_2 \oplus \ldots \oplus R_{univ}e_d$$

as an R_{univ}-module with algebra structure given by the rule

$$e_k e_l = \sum c^m_{kl} e_m$$

The closed immersion $Z_{univ} \to X_{T_{univ}}$ is given by the R_{univ}-algebra map

$$\Psi : A \otimes_R R_{univ} \longrightarrow B_{univ}$$

mapping $1 \otimes 1$ to $\sum b^l e_l$ and x_i to $\sum b^l_i e_l$ (for existence see below). As for the ideal $\mathfrak{a}_{univ}$ we have the following prescription:

(1) multiplication on B_{univ} is commutative, i.e., we should have $c^m_{lk} - c^m_{kl} \in \mathfrak{a}_{univ}$,
(2) multiplication on B_{univ} is associative, i.e., we should have $c^m_{lk}c^p_{mn} - c^p_{lq}c^q_{kn} \in \mathfrak{a}_{univ}$,
(3) $\sum b^l e_l$ is a multiplicative 1 in B_{univ}, in other words, we should have $(\sum b^l e_l)e_k = e_k$ for all k, which means $\sum b^l c^m_{lk} - \delta_{km} \in \mathfrak{a}_{univ}$ (Kronecker delta).

Since we've guaranteed that B_{univ} is a commutative R_{univ}-algebra with 1 given by $\sum b^l e_l$ the homomorphism Ψ exists. The final condition is that

(5) f_l to map to e_l in B_{univ}. Write $\Psi(f_l) - e_l \equiv \sum h_l^m e_m$ for some $h_l^m \in R[c_{kl}^m, b^l, b_i^l]$ then we should have $h_l^m \in \mathfrak{a}_{univ}$.

With $\mathfrak{a}_{univ} \subset R[c_{kl}^m, b^l, b_i^l]$ generated by the elements listed in (1) – (5) it is clear that F_W is represented by $\mathrm{Spec}(R_{univ})$. □